

**MICHIGAN DEPARTMENT OF NATURAL RESOURCES AND ENVIRONMENT
ENVIRONMENTAL LABORATORY**



P.O. Box 30270
Lansing, MI 48909
TEL: (517) 335-9800
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Division: WRD
Report to: CHRIS LITTLE
MDNRE-WRD-PSM
GRAND RAPIDS DISTRICT OFFICE
350 OTTAWA AVENUE, NW, GRAND RAPIDS, MI 49503

Lab Work Order # : 10100101
Work Site ID : 0020257
Site Name : CADILLAC WWTP
Received: 01/25/2011
Reported: 02/14/2011
Collected By: CHRIS LITTLE

Total: \$355.00

Samples Received :

No:	Sample ID	Sample Description	Matrix:	Collection Date
01	AB69960	CD-001A	WASTE WATER	01/25/2011
02	AB69961	CD-001A-1	WASTE WATER	01/24/2011
03	AB69962	CD-UP	WASTE WATER	01/24/2011
04	AB69963	CD-001A-F	WASTE WATER	01/25/2011

I certify that the analysis performed by the MDEQ Environmental Laboratory are accurate and that the laboratory tests were conducted by methods approved by the U.S. Environmental Protection Agency and other appropriate regulatory agencies.

George L. Krisztian,
Acting Laboratory Director

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Sample Number: AB69960 CD-001A

CAS#	Analyte Name	Result	Unit	RL	Qualifier	Date Tested	Method	Analyst
	Alkalinity (as CaCO ₃)	110	mg/L	20		02/04/2011	310.2	TK
7440-70-2	Calcium - Total	72.5	mg/L	1		02/09/2011	7140/3111B 5	EG
7439-95-4	Magnesium - Total	18.4	mg/L	1		02/09/2011	7450/242.1	EG
	Digestion Metals Water	Completed				02/01/2011	3010/200	WN
	Hardness - Calculated	257	mg/L	5		02/14/2011	Calculated	KS
	KN TP - Digestion	Completed				02/01/2011	351.2	LAV
7723-14-0	Total Phosphorus	.13	mg P/L	0.010		02/01/2011	365.4	LAV
7782-49-2	Selenium - Total	ND	µg/L	1		02/03/2011	6020/200.8	KS
7440-66-6	Zinc - Total	69	µg/L	10		02/03/2011	6020/200.8	KS
7664-41-7	Ammonia	ND	mg N/L	0.1	D	02/04/2011	350.1	MB
	Solids - Suspended	4	mg/L	4		01/28/2011	2540D SM	TK

Sample Number: AB69961 CD-001A-1

CAS#	Analyte Name	Result	Unit	RL	Qualifier	Date Tested	Method	Analyst
CN_AVAL	Cyanide - Available	0.006	mg/L	0.002		01/28/2011	OIA-1677	MB
7439-97-6	Low Level Mercury - Total	.8	ng/L	0.5		02/01/2011	1631	TK

Sample Number: AB69962 CD-UP

CAS#	Analyte Name	Result	Unit	RL	Qualifier	Date Tested	Method	Analyst
	Alkalinity (as CaCO ₃)	79	mg/L	20		02/04/2011	310.2	TK
7440-70-2	Calcium - Total	28.0	mg/L	1		02/09/2011	7140/3111B 5	EG
7439-95-4	Magnesium - Total	6.2	mg/L	1		02/09/2011	7450/242.1	EG
	Hardness - Calculated	96	mg/L	5		02/14/2011	Calculated	KS

Sample Number: AB69963 CD-001A-F

CAS#	Analyte Name	Result	Unit	RL	Qualifier	Date Tested	Method	Analyst
	Alkalinity (as CaCO ₃)	110	mg/L	20		02/04/2011	310.2	TK
	KN TP - Digestion	Completed				02/01/2011	351.2	LAV
7723-14-0	Total Phosphorus	.15	mg P/L	0.010		02/01/2011	365.4	LAV
7664-41-7	Ammonia	ND	mg N/L	0.1	D	02/04/2011	350.1	MB
	Solids - Suspended	5	mg/L	4		01/28/2011	2540D SM	TK

CAS# : Chemical Abstract Service Registry Number

RL : Reporting Limit

ND : Not Detected

ug / L : microgram / liter (ppb)

mg / L : milligram / liter (ppm)

ug / Kg : microgram / kilogram (ppb)

mg / Kg : milligram / kilogram (ppm)

Laboratory Contacts

Inorganic Unit Mgr: Sandy Gregg

Organic Unit Mgr: Carol Smith

Systems Mgmt Unit: George Krisztian

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Qualifier Code	Qualifier Description
1	Result(s) and RL(s) are estimated due to low surrogate recovery.
2	Result is estimated due to high surrogate recovery.
3	Result(s) and RL(s) are estimated due to low matrix spike recovery.
4	Result is estimated due to high matrix spike recovery.
5	Result and RL are estimated due to low continuing calibration standard criteria failure.
6	Result is estimated due to high continuing calibration standard criteria failure.
7	Result(s) and RL(s) are estimated due to poor precision.
8	Result(s) and RL(s) are estimated due to low recovery of batch QC.
9	Result outside QC acceptance criteria.
A	Value reported is the mean of two or more determinations.
C	Value calculated from other independent parameters.
D	Analyte value quantified from a dilution(s); reporting limit (RL) raised.
E	Result is estimated due to high recovery of batch QC.
F	Amenable cyanide was not analyzed due to low level of total cyanide.
G	Result and RL are estimated due to initial calibration standard criteria failure.
H	Recommended laboratory holding time was exceeded.
I	Dilution required due to matrix interference; reporting limit (RL) raised.
J	Analyte was positively identified. Value is an estimate.
JA	Result is estimated due to multiple Aroclors present.
JC	Result is estimated since confirmation analysis did not meet acceptance criteria
JD	Due to severe degradation, specific Aroclor identification is difficult and quantitation is estimated.
K	RL(s) raised due to matrix interferences.
KR	RL(s) raised due to low sample volume submitted.
KS	RL(s) raised due to low total solids.
KW	RL(s) raised due to light sample weight.
LB	Reported library search compounds are tentative identifications with estimated concentrations.
M	The level of the method preparation blank (MPB) is reported in the qualifier column.
N	Non-homogeneous sample made analysis of sample questionable.
O	Result and RL estimated due to analysis from an open vial.
P	Recommended sample collection/preservation technique not used; reported result(s) is an estimate.
Q	Quantity of sample insufficient to perform analyses requested.
R	Result confirmed by re-extraction and analysis.
S	Supernatant analyzed.
T	Reported value is less than the reporting limit (RL). Result is estimated.
V	Value not available due to dilution.
W	Reported value is less than the method detection limit (MDL).
X	Methods 8260 & 624 are used to analyze volatile organics that have boiling points below 200°C. 2-Methylnaphthalene & naphthalene have boiling points above 200°C and are better suited to analysis by methods 8270 or 625 as semivolatile organics.
PI	Possible interference may have affected the accuracy of the laboratory result
Z	Result reported below the RL to meet the TDL in RRD Op Memo 2 (10/22/04) multiplied by applicable dilution factor.

CAS# : Chemical Abstract Service Registry Number

RL : Reporting Limit

ND : Not Detected

ug / L : microgram / liter (ppb)

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